

Theory of Computation

Fall 2026

Programming assignment 1: finite automata constructions

Deadline: Friday 11/9/2026 at 23:59

Submission is through Gradescope. This is a programming assignment. Two programming assignments together can contribute **up to 5 points** to your grade.

Late submissions are not allowed, but you can submit (and get automated feedback) as many times as you want before the deadline.

This is a group assignment. Groups can be of 1 to 3 people. Your solutions must be your own work. In particular, external help or the use of AI tools is not allowed. You should be able to explain your own solutions.

Each component of the assignment is graded on a success/failure basis, and there is no partial credit. But you can try as many times as you want until you succeed or the deadline passes. Gradescope's autograder will give you feedback about the test that your solution failed.

Overview

In this programming assignment, you will use Python to **implement** two automata constructions. You must write functions that:

1. Determinize NFAs.
2. Minimize DFAs

It is important to note that the **proper implementation** of the required functions is crucial for this assignment, and **you must explain** how you have implemented the required constructions – **a missing or incomplete explanation can be grounds to not take your solution into account.**

The Constructions

NFA determinization refers to the process of constructing a DFA D from an NFA N , such that N and D accept the same language. We have discussed how to determinize NFAs in the course (see the subset construction). You need to implement that construction.

We have also discussed a DFA minimization algorithm, which you can implement.

Python implementations

The vocabulary we consider is always $\{0,1\}$. Assume there is a given list called `symbols` and is defined as `symbols = ['0', '1']`.

You are required to implement the following functions in your program:

1. `determinize(nfa)`

This function takes as input an NFA and returns a DFA that accepts the same language.

2. `minimize(dfa)`

This function takes as input a DFA and returns a DFA that accepts the same language and has the fewest possible states.

Constraints and Submission Format

Do not use any Python libraries or external packages. Your functions must be given in a single file called **automata.py**. You are provided with a template file that you can fill in. Use it as a reference to guide you. Make sure to include reasonable explanations and comments in your code to explain how your functions work. If there are no explanations of your code, it is possible that your solution will be disregarded.

Submission and feedback

You are required to submit your program in a single file called 'automata.py' via Gradescope by the deadline. Your code will be run through an autograder, which will provide feedback on the correctness of your code. The autograder runs many tests for each implementation.

You can submit your program multiple times before the deadline, and each submission will be graded by the autograder. This will allow you to fix any errors the autograder finds and resubmit your program for re-grading.

Please make sure to review the feedback provided by the autograder carefully and address any issues that it finds before resubmitting your program.

Please note that if you have any trouble with the autograder or submitting your assignment, don't hesitate to get in touch with us for help.

Grading breakdown

Each function's implementation is worth **2.5 points**. In **total**, the assignment is worth **5 points**.